

A cascade pathway model from the Nutrition Continuity work at Dark Matter Labs.

How one food-system shock becomes many crises.

A shock like a bad harvest or a fertiliser squeeze doesn't stay one problem. It splits across calories, protein and fresh produce, feeds back on itself through four loops, and ends in one of two places: a breakdown in public trust, or a hit to public finances. Turn shocks and interventions on and off, step through four time windows, and trace ten real risk pathways through the network below.

Type

Interactive model, qualitative

Reads

Pathways, loops, timing and windows

Updated

15 July 2026, version 2

How to read it

A shock is never just one problem. It's a set of pathways firing together, at different speeds, and feeding back on themselves — and this model lets you trace each one.

This is the detailed engine behind the worked [Camden Nutrition Risk Model ↗](#), drawn in the same visual language as [The Cascade Atlas ↗](#). Four things make it specific:

- **A detailed network.** Protein splits into three real groups — eggs & poultry, farmed fish, red meat & dairy — each falling at its own pace. The social side opens into seven early-warning signs of a system under strain; the financial side splits into three related risks (sticky prices, a shrinking budget, and an unhedged cost).
- **Time matters.** Step through four windows — winter access, pre-harvest import reliance, rising input costs, and the long tail — to see which pathways are live in each one.

- **Real, current figures.** Click any node for what it means, the number behind it, its source, and how fast it takes effect.
- **Ten named risk stories.** Pick one and it lights up its exact route through the network, with each stage's timing, the type of risk it carries, the early-warning sign to watch, and the fix that interrupts it.

This is a qualitative model, not a forecast. The shade of each node and the two gauges show relative pressure, from 0 to 100 — not real probabilities or dates. They show how the system is shaped, not what will happen.

The model

2026 shocks. Select one to turn it on or off.

El Niño

Fertiliser / energy

Hormuz / trade

H5N1 / protein

UK import dep.

Interventions from the portfolio, generalised

Forward demand books

Civic reserve

Protein options / fermentation

Affordability stabilisers

Shock test + provision floor

Price continuity (negative book)

Volatility window

All windows

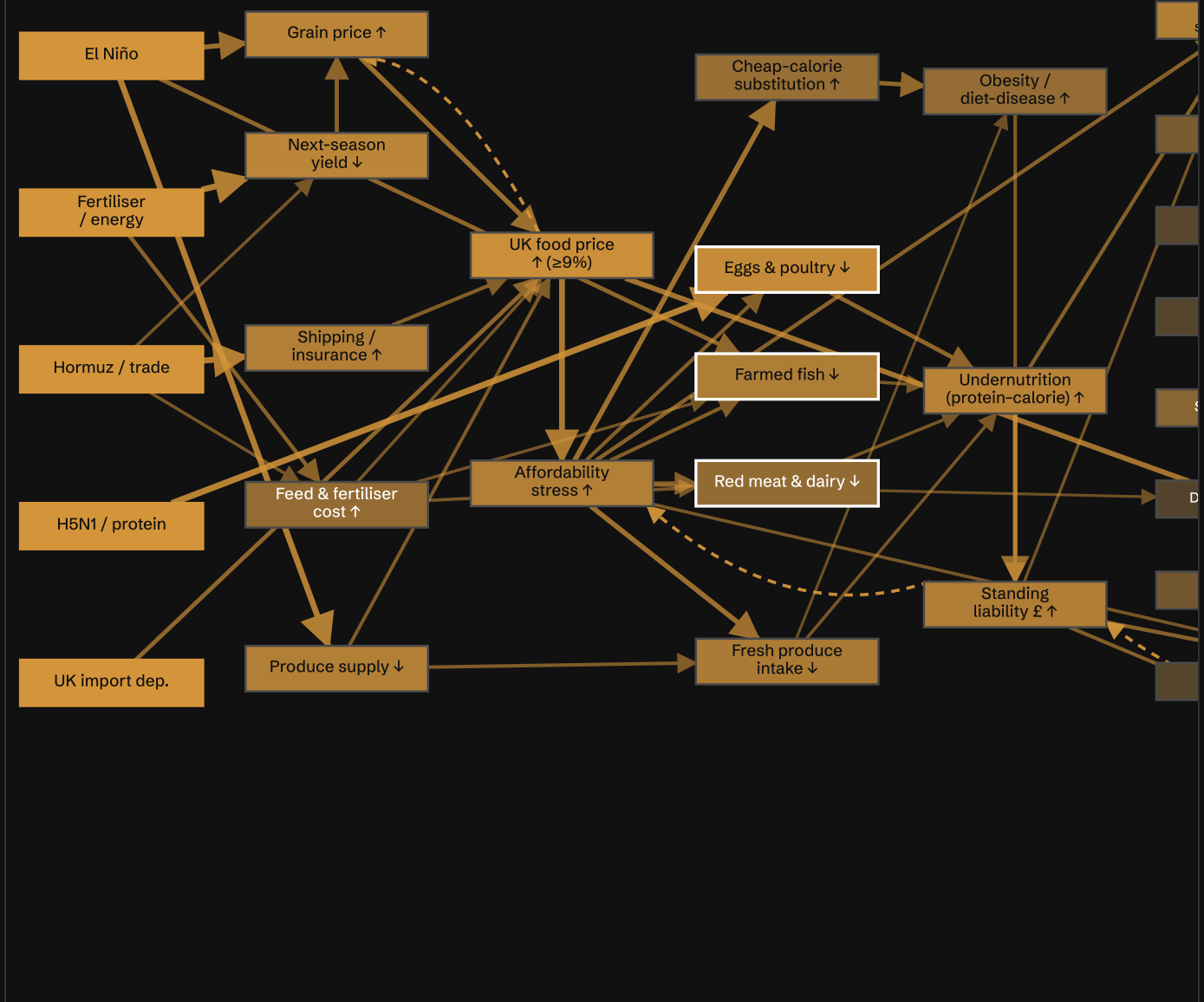
W1 winter access

W2 import dependency

W3 input-price reset

W4 compound tail

Next window



■ Shock (on)
 ■ Pathway pressure, low to high
 Protein sub-chain
 ○ Terminal risk

Terminal A: social-contract breakdown

63%

Terminal B: financial instability

73%

L1 protectionism, active

L2 affordability to health, active

L3 fiscal ratchet, active

L4 legitimacy, active

Cascade library

Ten risk cascades

Select one to light its exact path on the network and read its staged dossier — each stage with its time-lag, topology, the early-warning signal, and the intervention that cuts it. Or click any node on the map for its mechanism, figure and lag.

K1 Fertiliser shock → thin harvest → calorie cost
input → calories

K2 El Niño multi-breadbasket synchronisation
climate → price

K3 Hormuz → input cost → delayed universal price
chokepoint → price

K4 H5N1 → egg/poultry protein → undernutrition
disease → protein

K5 Affordability → protein dropped first → undernutrition
budget → protein

K6 Affordability → cheap-calorie substitution → obesity pipeline
budget → calories

K7 Provision strain → social tipping → breakdown
provision → legitimacy

K8 Food inflation → sticky inflation → regressive tightening
monetary trap

K9 Fiscal ratchet — reactive crowds out preventive
local fiscal

K10 The unpriced under-hedge
Value-at-Risk

The four loops

L1 protectionism

Price → hoarding → price

A price spike triggers hoarding and export bans, which thins the market and pushes the next price even higher.

L2 affordability to health

Cost → deficit → cost

When people trade down to cheaper food, diet-related illness and malnutrition both rise — adding to the health bill and leaving less room to fix affordability.

L3 fiscal ratchet

Reactive crowds out preventive

Money spent reacting to a crisis leaves less to prevent the next one — so vulnerability, and next year's bill, keep growing.

L4 legitimacy

Strain → distrust → worse response

When services buckle, people trust institutions less — which makes the next response weaker still.

Honest limits

- **Qualitative, not quantitative.** The weights are illustrative, and the gauges show relative pressure — not real probabilities or dated forecasts.
- **The detail is a simplified selection.** Real cascades have more nodes and connections; this keeps only the ones that carry the most weight, not a full census.
- **The time windows and lags are illustrative.** They show rough order and speed, not calibrated timing.
- **It inherits the caveats of its sources.** The 2026 figures are a mid-year snapshot; the Camden cost estimates are order-of-magnitude; the H5N1 figure is a US signal used for illustration.

From the Civilisational Optionality work at Dark Matter Labs

The detailed engine behind [the Camden Nutrition Risk Model ↗](#) and the geographic [Supply Map](#), in the same visual language as [The Cascade Atlas ↗](#). Built on public data — ONS, Camden JSNA, BAPEN, Frontier Economics, WMO, World Bank, FAO, FDF. A model for orientation, not a forecast: the weights are illustrative, not fitted, and this is interpretation, not a primary source.

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